DATA SCIENCE LABORATORY MANUAL

Module II ? Python for Data Science

Topic: Data Manipulation

OVERVIEW & LEARNING OBJECTIVES

Aggregations with groupby, pivot tables, multi-indexing, and reshaping with melt and stack.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Data Manipulation in modern Data Science workflows.
- Detailed breakdown: Aggregations with groupby, pivot tables, multi-indexing, and reshaping with melt and stack.
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.